Chapter 2

Coin flips → 5 coins

H H T T H ← microstate

T H H H H ← microstate

different microstate → H H H T T

3 heads ← macrostate

how many microstates are in a macrostate?

↳ multiplicity $\Omega(N,n) = \frac{N!}{n!(N-n)!}$

(n ← # of heads)

N ← total number of coins

n ← number of heads

$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

Ex: 5 coins

$$\Omega(5) = \frac{5!}{5!(5-5)!} = \frac{5!}{5!\,0!} = 1$$

$0! = 1$

$$\Omega(4) = \frac{5!}{4!\,(5-4)!} = \frac{5 \cdot \cancel{4 \cdot 3 \cdot 2 \cdot 1}}{\cancel{4 \cdot 3 \cdot 2 \cdot 1}} = 5$$

$1! = 1$

$x! = x \cdot (x-1)!$

$$\Omega(3) = \frac{5!}{3!(5-3)!} = \frac{5!}{3!\,2!} = \frac{120}{6 \cdot 2} = 10$$

10 atoms - each atom can have either 0 or 1 energy units.

So, how many combinations are there of arranging 4 units of energy?

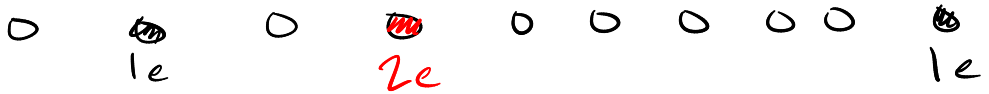
$$\Omega(N,n) = \frac{N!}{n!(N-n)!}$$

What if more than one energy can be on each atom?

$$\Omega(N,q) = \frac{(q+N-1)!}{q!(N-1)!}$$

N atoms

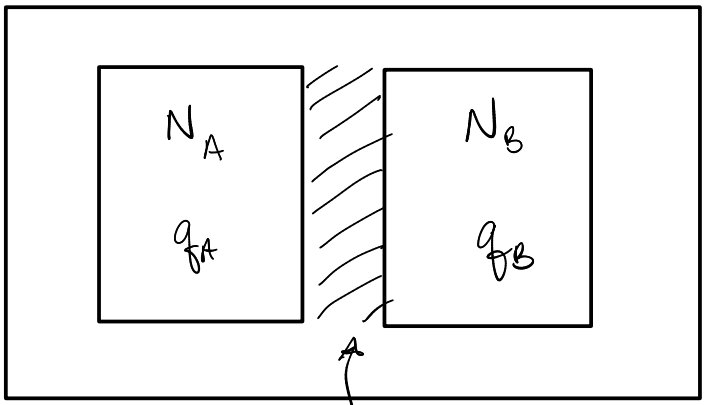
q: number of energy packets to share



← one particular microstate in the macrostate of 4 energies among 10 atoms

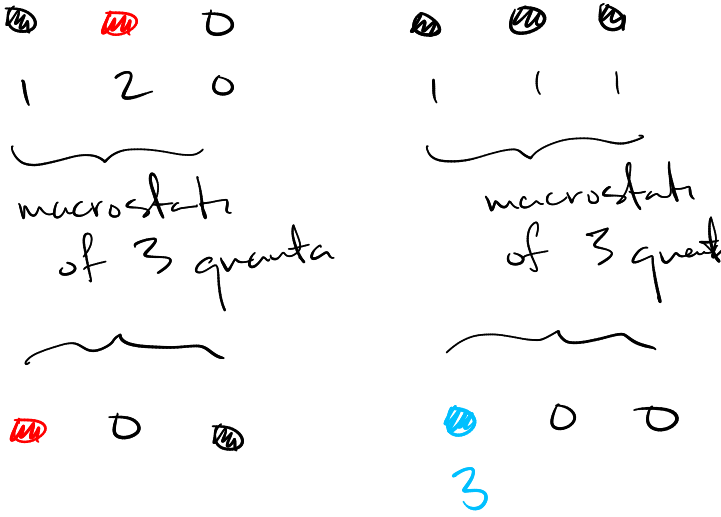
This model of how materials behave is known as an Einstein Solid.

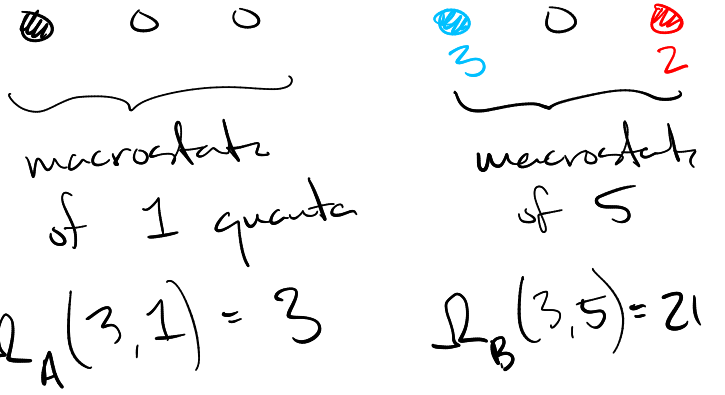
Two Systems – these can exchange energy



isolated from surroundings

energy can flow from one side to the other, but not particles





$$\Omega_A(3,1) = 3 \qquad \Omega_B(3,5) = 21$$

what is the total multiplicity of these macrostates?

$$\Omega_{total} = \Omega_A \cdot \Omega_B$$

$$= 63$$

$$\Omega(N,q) = \frac{(q+N-1)!}{q!\,(N-1)!}$$

$$= \frac{(5+3-1)!}{5!\,(3-1)!} = 21$$

↑ what is this the same as?

$$\Omega(6,6) = \frac{(6+6-1)!}{6!\,(6-1)!}$$

$$= \frac{11!}{6!\,5!} = 462$$

I had some extra information before!

Fundamental Assumption of Statistical Mechanics

↳ all microstates are possible and equally probable

But. That does not mean that every microstate will occur.

Not all macrostates are equally probable.

Large Number → addition of small #'s does not matter

$10^{23} + 100 = 10^{23}$

Very Large Numbers → multiplication of small or large #'s does not matter

$10^{10^{23}} \times 10^{23} = 10^{10^{23} + 10^{23}} \approx 10^{10^{23}}$

$$\Omega(N, q) = \frac{(q+N-1)!}{q!\,(N-1)!} \approx \frac{(q+N)!}{q!\,N!}$$

$$\ln \Omega = \ln[(q+N)!] - \ln(q!) - \ln(N!)$$

$\ln(N!) \to N\ln N - N$

$\ln(q!) \to q\ln q - q$

$\ln[(q+N)!] \to (q+N)\ln(q+N) - (q+N)$

Stirling's Approximation

$$N! \approx N^N e^{-N} \cdot \sqrt{2\pi N} \approx N^N e^{-N} N^{1/2}$$

$$\approx N^{N+\frac{1}{2}} e^{-N}$$

$$\boxed{N! \approx N^N e^{-N} = \frac{N^N}{e^N}}$$

$$\ln N! \approx \ln(N^N) - \ln(e^N)$$

$$\boxed{\ln N! \approx N\ln N - N}$$

$$\ln \Omega = (q+N)\ln(q+N) - q - N - q\ln q + q - N\ln N + N$$

$$\boxed{\ln \Omega = (q+N)\ln(q+N) - q\ln q - N\ln N}$$

what is Ω in the high temperature limit?

$$q >> N$$

$$\ln \Omega = (q+N)\ln(q+N) - q\ln q - N\ln N$$

$$= q\ln(q+N) + N\ln(q+N) - q\ln q - N\ln N$$

$$= \ln\left(q\left(1+\frac{N}{q}\right)\right)$$

$$= \ln q + \ln\left(1+\frac{N}{q}\right)$$

$\ln\left(1+\frac{N}{q}\right)$ → a lot like $\ln(1+x) \approx x$

$$= \frac{N}{q}$$

$$= \ln q + \frac{N}{q} \approx \ln(q+N)$$

$$\ln\Omega = q\ln q + q\cdot\frac{N}{q} + N\ln q + \frac{N^2}{q} - q\ln q - N\ln N$$

$$\ln\Omega = N\ln q - N\ln N + N + \frac{N^2}{q}$$

$$= N\ln\left(\frac{q}{N}\right) + N + \underbrace{\frac{N^2}{q}}_{\text{small}}$$

$$\ln\Omega(q>>N) = N\ln\left(\frac{q}{N}\right) + N$$

$$\Omega(q>>N) = e^{N\cdot\ln\left(\frac{q}{N}\right)+N} = e^{N\cdot\ln\left(\frac{q}{N}\right)}\cdot e^{N}$$

$$= \left[e^{\ln\left(\frac{q}{N}\right)}\right]^N \cdot e^N$$

$$= \left(\frac{q}{N}\right)^N \cdot e^N$$

$$\boxed{\Omega(q>>N) = \left(\frac{e\cdot q}{N}\right)^N}$$

← Multiplicity of an Einstein solid at high temperature

work problem 2.18.

2.22.

Entropy

Any large system in equilibrium will be found in the macrostate w/ the greatest multiplicity.

2nd law of Thermodynamics

Multiplicity tends to increase.

$$S = k_B \cdot \ln \Omega$$

Entropy → the log of the number of ways of arranging our system.

Ex: an Einstein solid at high temp^{c}. $q >> N$

$N = 10^{23}$ $\quad q = 10^{25}$

$$\Omega(q >> N) = \left(\frac{e \cdot q}{N}\right)^N$$

$$S = k_B \ln\left[\left(\frac{e \cdot q}{N}\right)^N\right]$$

$$S = k_B \cdot N \cdot \ln\left(\frac{e \cdot q}{N}\right)$$

$$= k_B \cdot N \cdot \left(\ln\left(\frac{q}{N}\right) + \underbrace{\ln(e)}_{1}\right)$$

$$\boxed{S = k_B \cdot N \cdot \left(\ln\left(\frac{q}{N}\right) + 1\right)}$$

$N = 10^{23}$ $\quad q = 10^{25}$

$$S = 1.38 \cdot 10^{-23} \frac{J}{K} \cdot 10^{23} \cdot \left(\ln\left(\frac{10^{25}}{10^{23}}\right) + 1\right)$$

$$S = 1.38 \frac{J}{K} \cdot (4.6 + 1)$$

$$\boxed{S = 7.7 \frac{J}{K}}$$

For a composite system, $\Omega_{total} = \Omega_A \cdot \Omega_B$

$$S_{total} = k_B \ln \Omega_{total} = k_B \ln \Omega_A \Omega_B$$

$$= k_B \ln \Omega_A + k_B \ln \Omega_B$$

$$S_{total} = S_A + S_B$$

Any large system in equilibrium will be found in the macrostate w/ the greatest ~~multiplicity~~ entropy.

2nd law of Thermodynamics

~~Multiplicity~~ Entropy tends to increase.

→ spontaneous process

2.29 + 2.30

Total HW: 2.18, 22, 29, 30 Due: Wednesday October 7th

- grade tests
- post solutions to optional HW

Chapter 3 → defin temperature → thermal equilibrium

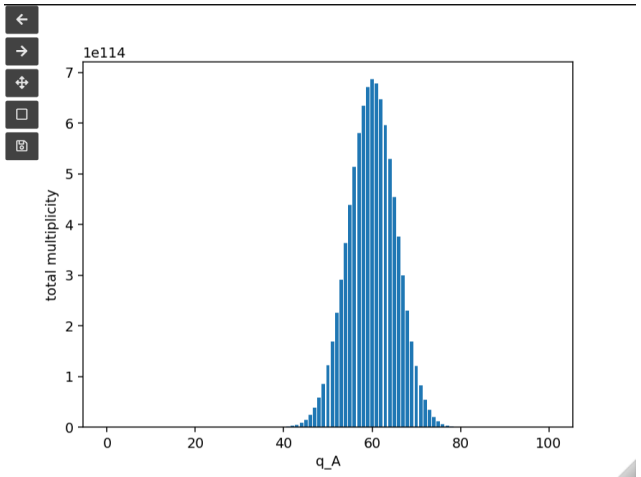
For an Einstein solid

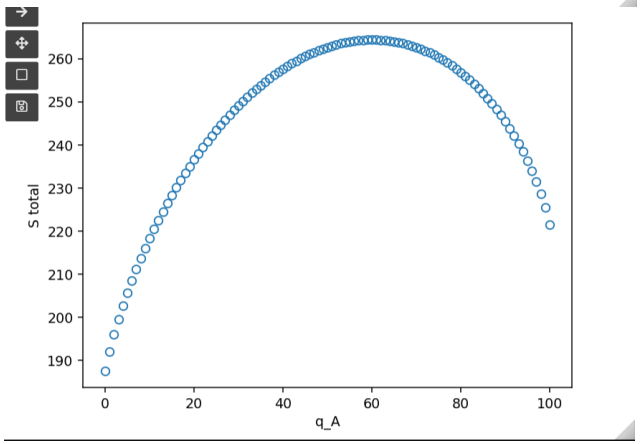
$$\frac{\partial S_{total}}{\partial q_A} = 0$$

$$\frac{\partial S_{total}}{\partial U_A} = 0$$

$$S_{total} = S_A + S_B$$

$$\frac{\partial (S_A + S_B)}{\partial U_A} = 0$$





$$\frac{\partial S_A}{\partial U_A} + \frac{\partial S_B}{\partial U_A} = 0$$

$$U_A + U_B = U$$

$$U_B = U - U_A$$

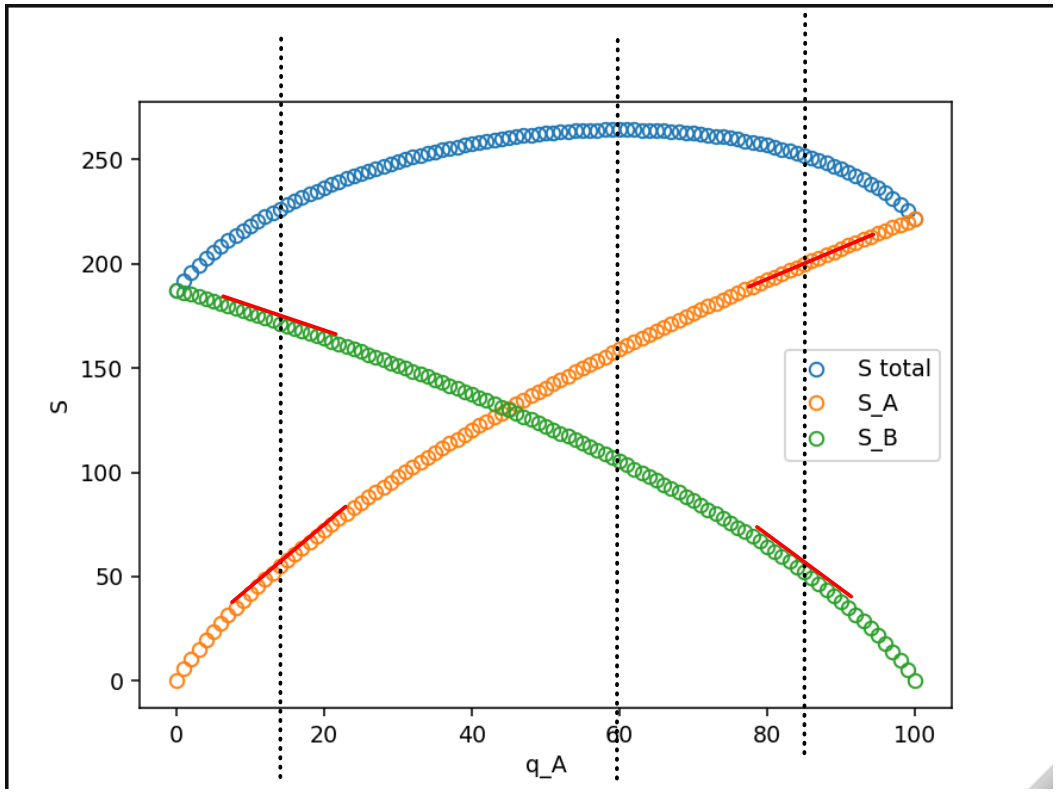
$$dU_B = -dU_A$$

$$\frac{\partial S_A}{\partial U_A} - \frac{\partial S_B}{\partial U_B} = 0$$

$$\frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V}$$

$$T = \left(\frac{\partial S}{\partial U}\right)^{-1}_{N,V}$$



↑ energy at equilibrium

Let's apply what we have learned to an Einstein solid at high temp:

$$\Omega(q>>N) = \left(\frac{eq}{N}\right)^N$$

$$U = q \cdot \epsilon$$

q: # of energy packets; ϵ: energy of each packet

$$S = k_B \ln \Omega$$

$$S = N k_B \ln\left(\frac{eq}{N}\right)$$

$$S = N k_B \left(1 + \ln\left(\frac{q}{N}\right)\right)$$

$$S = N k_B \left(1 + \ln\left(\frac{U}{\epsilon N}\right)\right)$$

$$= N k_B + N k_B \ln(U) - N k_B \ln(\epsilon N)$$

$$\frac{1}{T} = \left(\frac{dS}{dU}\right)_{NV} = \frac{N k_B}{U} \Rightarrow T = \frac{U}{N k_B} \Rightarrow \boxed{U = N k_B T}$$

U = Power · time

measure the temp

Very naive prediction

$$U = \frac{f}{2} \cdot N \cdot k_B \cdot T$$

$f = 2$

Einstein solid

1D kinetic

1D spring potential energy

$$\boxed{U = N k_B T}$$

$$T = \frac{U}{N k_B}$$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V}$$

$$P = I \cdot V$$
$$= \frac{V^2}{R} = I^2 \cdot R$$

So let's review our process so far:

1. Found Ω in terms of N, U, V, P } could be impossible!
   - We used combinatorics, approximations, QM }

2. $S = k_B \ln \Omega$ to get entropy

3. $\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V}$

4. solve step 3 equation for U as a function of T ← statistical mechanics gives a alternative approach to arriving at step 4.

5. $C_V = \frac{\partial U}{\partial T}$

2.8

macrostates → # of energies 20 total + 1 empty state = 21 macrostates

Ω ← number of microstates in 1 particular macrostate

$$\Omega(N,q) = \frac{(q+N-1)!}{q!\,(N-1)!}$$

$$\frac{(20+20-1)!}{20!\,(19)!} = \frac{39!}{20!\,19!} = 6.9\cdot10^{10}$$

total $N = 20$ total $q = 20$

all energy in 10 atoms

$$\frac{(20+10-1)!}{20!\,(9)!} = 2.3\cdot10^{22}$$

$$\text{probability} = \frac{\#\text{ in one state}}{\text{all the \# in all states}} = \frac{\Omega(10,20)\cdot\Omega(10,0)}{\Omega_{All}(20,20)} = \frac{1\cdot10^{7}}{6.9\cdot10^{10}} = 1.45\cdot10^{-4}$$

($\Omega(10,20)$: 1 way; $\Omega(10,0)$: 1 way)

↑ all in A

$$\frac{(10+10-1)!}{10!\,9!} = \frac{19!}{10!\,9!}$$

$$\text{probability} = \frac{\Omega(10,10)\,\Omega(10,10)}{6.9\cdot10^{10}} = \frac{8.5\cdot10^{9}}{6.9\cdot10^{10}} = 0.123 \checkmark$$

↑ half in A

0.124

↑ probability

2.5 Ideal Gas

Multiplicity - number of microstates in a macrostate

↳ total energy, volume

Ω = distribution of particles in space · distribution of energy among those particles

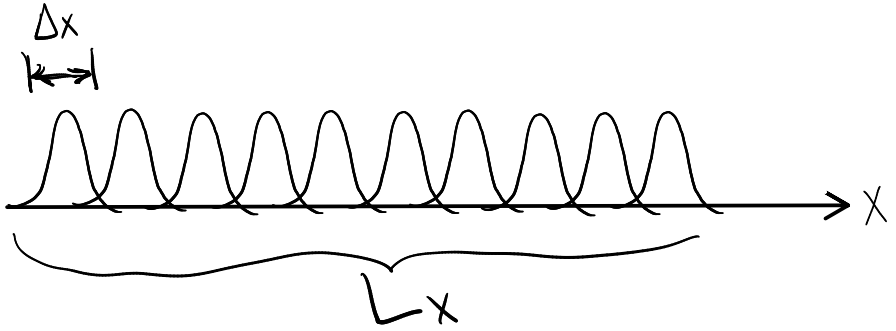
For a single particle in volume V

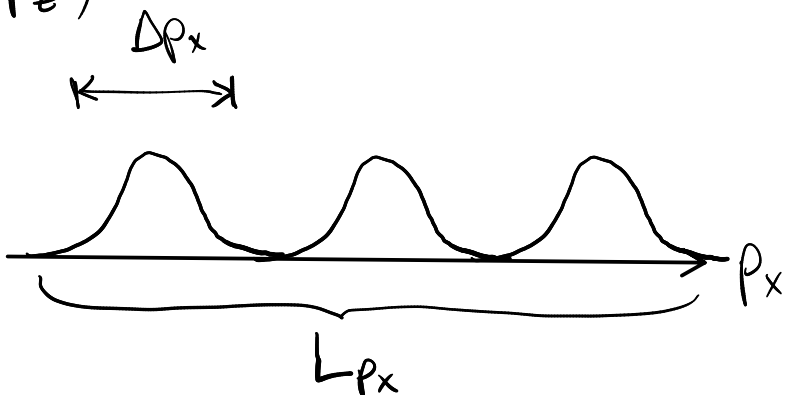
$\Omega_1 \propto V \cdot$ combinations of momentums that have the same energy

V: volume of physical space

volume (surface area) of momentum space

$$\Omega_1 \propto \left(\frac{L_x}{\Delta x} \cdot \frac{L_y}{\Delta y} \cdot \frac{L_z}{\Delta z}\right) \cdot \left(\frac{L_{p_x} \cdot L_{p_y} \cdot L_{p_z}}{\Delta p_x \cdot \Delta p_y \cdot \Delta p_z}\right)$$





Heisenberg Uncertainty Principle

$$\Delta x \cdot \Delta p_x \approx h$$ ← Planck constant

$h = 6.63 \cdot 10^{-34}\ m^2 \frac{kg}{s}$

Let's go back to one particle.

$$U = \frac{1}{2} m \vec{v}^2 = \frac{1}{2} m \left(v_x^2 + v_y^2 + v_z^2\right)$$

$$U = \frac{1}{2m}\left(p_x^2 + p_y^2 + p_z^2\right)$$

$$p_x^2 + p_y^2 + p_z^2 = 2mU$$

$$p_x^2 + p_y^2 + p_z^2 = \left(\sqrt{2mU}\right)^2$$

this is a constant!

equation of a sphere in momentum space

$$r_p = \sqrt{2mU}$$

$$\Omega_1 = \frac{V \cdot V_p}{h^3}$$

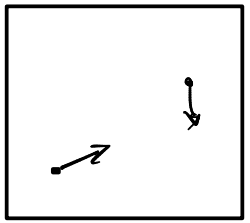
So, what about two particle?

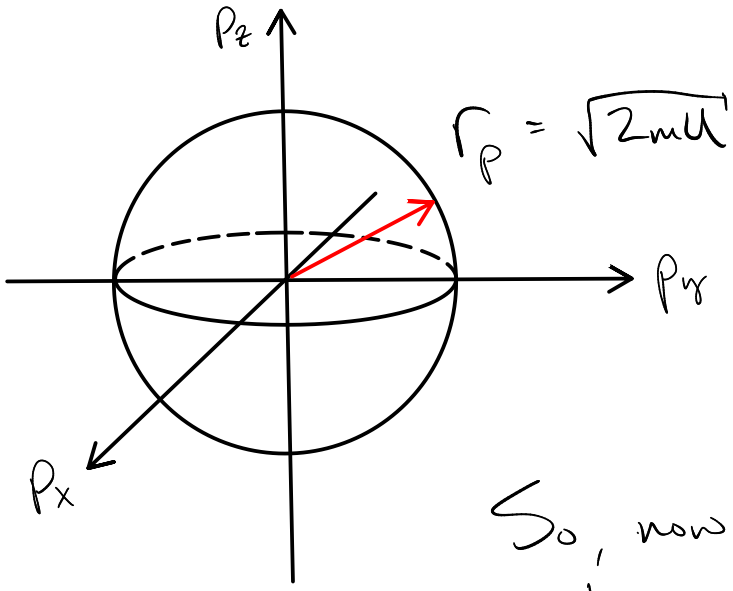
$$\Omega_2 = \Omega_A \cdot \Omega_B$$ ← since we can not distinguish the particles.

$$\Omega_2 = \frac{V^2 \cdot V_p^2}{(h^3)^2} \cdot \frac{1}{2}$$

So, what about N particles?

$$\Omega_N = \frac{V^N \cdot V_p^N}{(h^3)^N} \cdot \frac{1}{N!}$$





$r_p = \sqrt{2mU}$

The energy of our one particle is the surface area of this sphere.

↳ $4\pi r^2 \rightarrow 4\pi\left(\sqrt{2mU}\right)^2$

So, now, let's do two particles which have a total energy of U

$$p_{1x}^2 + p_{1y}^2 + p_{1z}^2 + p_{2x}^2 + p_{2y}^2 + p_{2z}^2 = \left(\sqrt{2mU}\right)^2$$

↳ total mass

↳ 6 dimensional sphere? ⇒ hypersphere

"surface area" $= \dfrac{2\pi^{d/2} \cdot r^{d-1}}{\left(\frac{d}{2}-1\right)!}$ ⟶ TEST: $d=3$

$4\pi r^2 \stackrel{?}{=} \dfrac{2\pi^{3/2} r^{3-1}}{\left(\frac{3}{2}-1\right)!}$

$= 2\left[\dfrac{\pi^{3/2}}{(1/2)!}\right] r^2$

so for 2 particles: ⇒

$$\Omega(N=2) = \frac{V^N \cdot V_p^N}{(h^3)^{N!}} = \frac{V^2 \cdot 4\pi\left(\sqrt{2mU}\right)^2}{(h^3)^2} \cdot \frac{1}{2!}$$

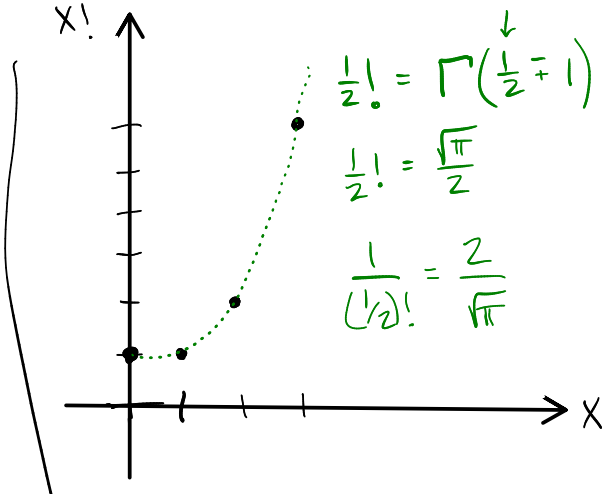
Now for N particles ↴

$$\Omega_N = \frac{1}{N!} \cdot \frac{V^N}{h^{3N}} \cdot \frac{2\pi^{\frac{3N}{2}} \cdot (2mU)^{\frac{3N}{2}-1}}{\left(\frac{3N}{2}-1\right)!}$$

Let's throw away what does not matter when N is large!

$$\Omega_N = \frac{1}{N!} \cdot \frac{V^N}{h^{3N}} \cdot \frac{\pi^{3N/2} \cdot (2mU)^{\frac{3N}{2}}}{\left(\frac{3N}{2}\right)!}$$

$$\Omega_N = f(N) \cdot V^N \cdot U^{\frac{3N}{2}}$$



$$= 2\left(\frac{\pi^{3/2} \cdot 2}{\pi^{1/2}}\right) \cdot r^2$$

$4\pi r^2 = 4\pi r^2$ ✓

TEST: $d = 2$

$$= \frac{2\pi^{2/2} \cdot r^{2-1}}{\left(\frac{2}{2}-1\right)!}$$

$$= 2\pi r$$

Entropy of an ideal gas:

$$S = k_B \ln \Omega$$

$$S = k_B \ln \left(\frac{1}{N!} \cdot \frac{V^N}{h^{3N \cdot \frac{2}{2}}} \cdot \frac{\pi^{3N/2} \cdot (2mU)^{\frac{3N}{2}}}{\left(\frac{3N}{2}\right)!} \right)$$

$$S = k_B \left[-\ln N! - \ln\left(\left(\frac{3N}{2}\right)!\right) + N \ln V + \frac{3N}{2} \ln(2\pi m U) - \frac{3N}{2} \ln(h^2) \right]$$

$$-\ln N! = -(N \ln N - N)$$

$$= -N \ln N + N$$

$$= N \ln \frac{1}{N} + N$$

$$-\frac{3N}{2} \ln\left(\frac{3N}{2}\right) + \frac{3N}{2}$$

$$\frac{3N}{2} \ln\left(\frac{2}{3N}\right) + \frac{3}{2}N$$

$$S = k_B N \left(\ln \left[\frac{V}{N} \left(\frac{2\pi m U}{h^2} \cdot \frac{2}{3N} \right)^{3/2} \right] + \frac{3}{2} + 1 \right)$$

$$\boxed{S = k_B N \left(\ln \left[\frac{V}{N} \cdot \left(\frac{4\pi m U}{3Nh^2} \right)^{3/2} \right] + \frac{5}{2} \right)}$$

→ Sackur-Tetrode Equation

Apply this to one mole of He at room temp + atmospheric pressure:

$n = 1$ mole $\quad N = N_A = 6.022 \cdot 10^{23}$ particles

monatomic

$$U = \frac{3}{2} N k_B T = \frac{3}{2} n R T = \frac{3}{2} P V$$

$$V = \frac{N k_B T}{P} = \frac{n R T}{P}$$

$$m_{He} = 4 \frac{g}{mol} = 0.004 \frac{kg}{mol} \cdot \frac{1\ mol}{6.022 \cdot 10^{23}\ particles} = 6.64 \cdot 10^{-27} \frac{kg}{particle}$$

$$S = k_B N_A \cdot \left(\ln\left[\frac{N_A k_B T}{P \cdot N_A} \left(\frac{2}{4\pi} \cdot \frac{6.64 \cdot 10^{-27} \cdot \frac{3}{2} N_A k_B \cdot 293}{3 \cdot N_A \cdot (6.626 \cdot 10^{-34})^2} \right)^{3/2} \right] + \frac{5}{2} \right)$$

$k_B N_A = 8.31$

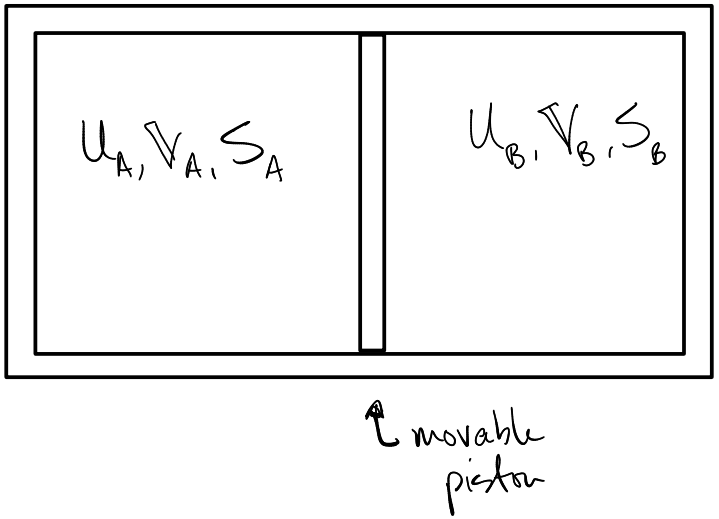
$$S = 126 \frac{J}{K}$$

3.4 Mechanical Equilibrium → Pressure

→ Exchange of energy is governed by temperature

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V}$$

→ Exchange of volume is governed by pressure differnce



equilibrium $\begin{cases} \dfrac{\partial S_{total}}{\partial U} = 0 \\ \text{definition of A+B being at the same temp} \end{cases}$

proposal: $\dfrac{\partial S_{total}}{\partial V_A} = 0$ $\qquad S_{total} = S_A + S_B$

$$\frac{\partial (S_A + S_B)}{\partial V_A} = 0$$

$$\frac{\partial S_A}{\partial V_A} + \frac{\partial S_B}{\partial V_A} = 0 \qquad dV_A = -dV_B$$

$$\frac{\partial S_A}{\partial V_A} - \frac{\partial S_B}{\partial V_B} = 0$$

$$\frac{\partial S_A}{\partial V_A} = \frac{\partial S_B}{\partial V_B}$$

At equilibrium, pressure must be the same. b/c otherwise the movable wall would move.

$$\boxed{p = T\left(\frac{\partial S}{\partial V}\right)_{U,N}}$$

$$\frac{J}{Km^3} = \frac{N\cancel{m}}{Km^{3\cancel{2}}} = \frac{Pa}{K}$$

How can we test this?

$$\Omega = f(N) \cdot V^N \cdot U^{\frac{3}{2}N}$$

$$S = k_B \ln \Omega = k_B\left(\ln(f(N)) + N \ln V + \frac{3}{2}N \cdot \ln U\right)$$

$$p = T\left(\frac{\partial S}{\partial V}\right)_{U,N} = k_B T\left(N\frac{\partial(\ln V)}{\partial V}\right)$$

$$p = Nk_BT \cdot \frac{1}{V}$$

$$pV = Nk_BT \leftarrow$$

3.1, 10, 14, 24, 25

Exam 1: 10/27 → 10/30
(noon) (midnight)

Measuring Entropy

$$\frac{1}{T} = \frac{dS}{dU}$$

$$dS = \frac{dU}{T} \qquad dU = \pm Q$$

$$dS = \frac{\pm Q}{T} \qquad \pm Q = C_v\,dT$$

← heat capacity not specific heat

$$dS = \frac{C_v\,dT}{T}$$

$$\int dS = \int \frac{C_v\,dT}{T}$$

$$S_f - S_i = \int \frac{C_V dT}{T}$$

$$S(T=0) = 0$$

$$\Delta S = (\Delta S)_V + (\Delta S)_U$$

$$dS = \left(\frac{\partial S}{\partial U}\right)_V dU + \left(\frac{\partial S}{\partial V}\right)_U dV$$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_V \qquad \frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_U$$

$$dS = \frac{1}{T} dU + \frac{P}{T} dV$$

$$TdS = dU + PdV$$

$$\boxed{dU = TdS - PdV}$$

Memorize this
↳ b/c you can use this to recover the formulas for T & P.

↳ Thermodynamic Identity
- true for any infinitesimal change as long as T & P are the same throughout the material

→ compare to the first law of thermo:

$$dU = đQ + đW$$

$$đQ \stackrel{?}{=} TdS \qquad đW \stackrel{?}{=} -pdV$$

- dV happens quasistatically
- no non-compression work (like an electric/magnetic field)
- no other relevant variables change (like N)

$$đQ = TdS \qquad đW = -pdV$$